When Shift Happens — Confounding is to Blame



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Acknowledgment



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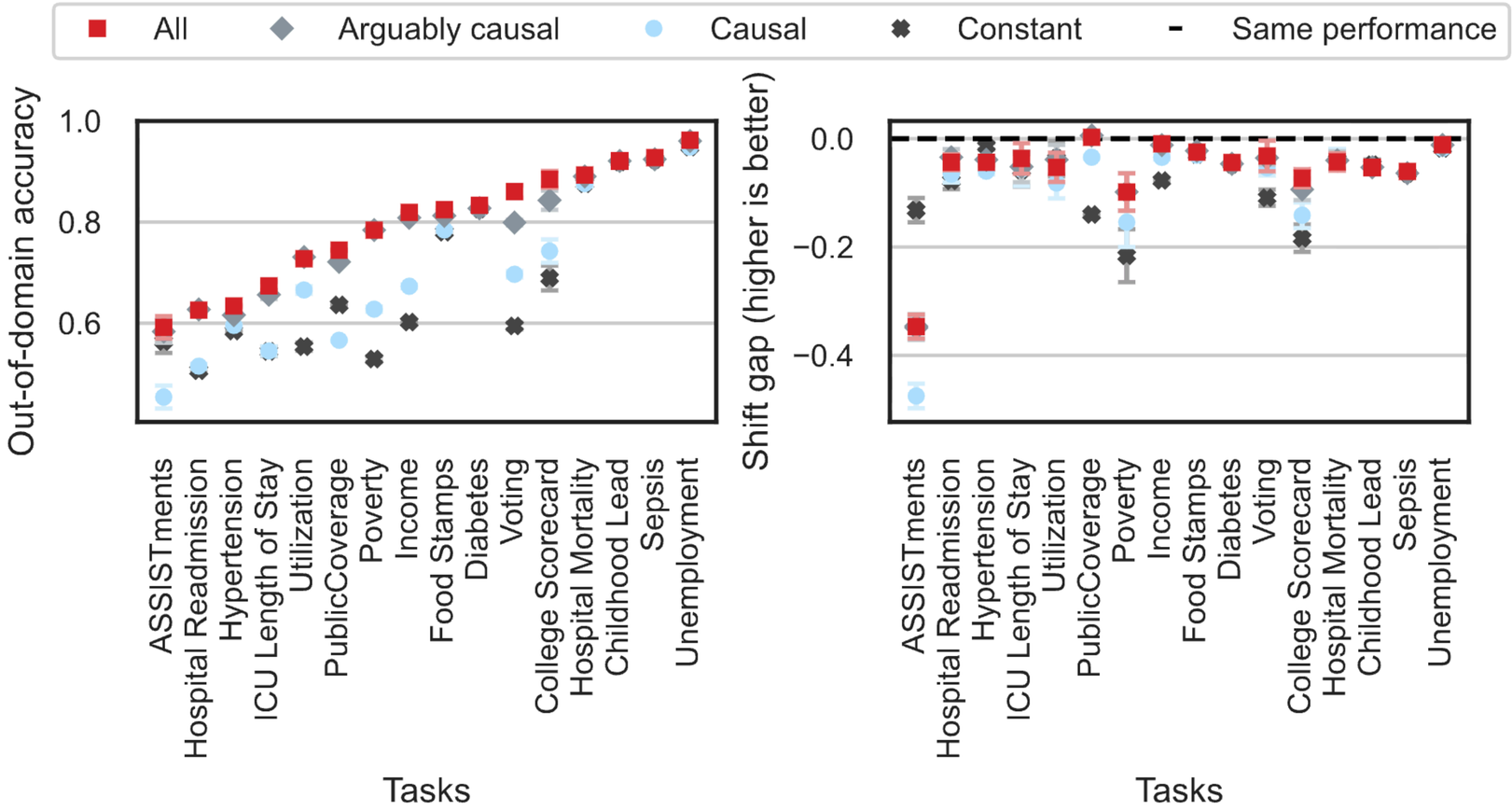


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[1] Gowtham, Celia, Rebekka, Krikamol. **When Shift Happens — Confounding is to Blame**. To appear in ICLR 2026.
[2] Gowtham, Rajeev, Celia, Krikamol, Rebekka. **Boosting for Predictive Sufficiency**. To appear in ICLR 2026.

Do causal predictors generalize better to new domains?

Causal (C) ⊆ Arguably Causal (AC) ⊆ All (A)

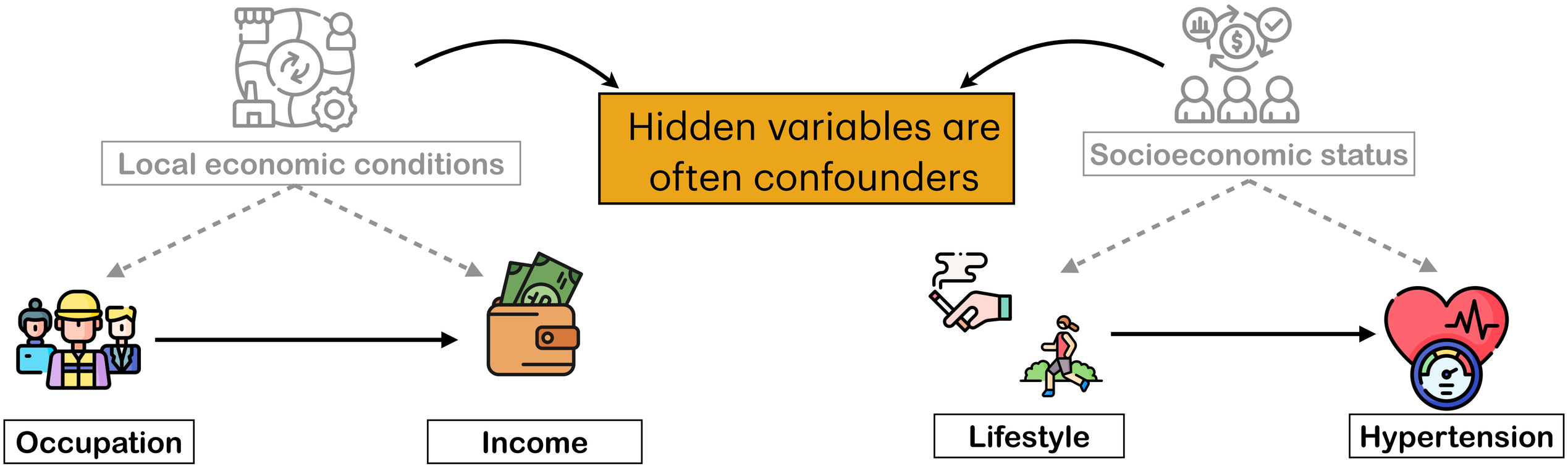


Finding: Predictors using all available features, regardless of causality, have better in-domain and out-of-domain accuracy than predictors using causal features.

Nastl, Vivian, and Moritz Hardt. "**Do causal predictors generalize better to new domains?**." *NeurIPS* 2024.

When shift happens, who is to blame?

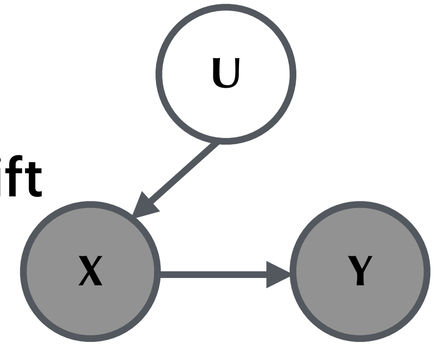
(Hidden) Confounding Shift



Train–test shifts often result from changes in hidden confounders.

Distribution Shifts and Invariance

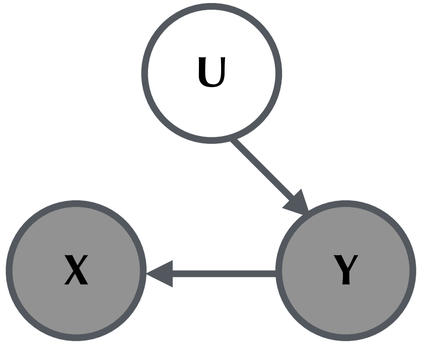
Covariate shift



$$\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$$
$$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$$
Invariance: $\mathbb{P}^e(Y \mid \mathbf{X}) = \mathbb{P}^{e'}(Y \mid \mathbf{X})$

Can be learned from observed data

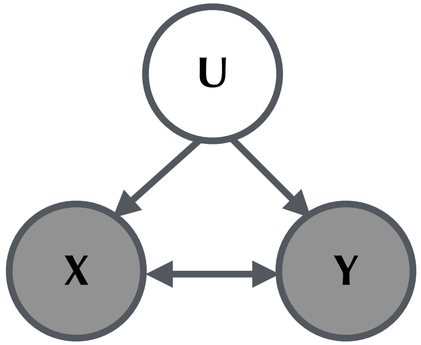
Label shift



$$\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$$
$$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$$
Invariance: $\mathbb{P}^e(\mathbf{X} \mid Y) = \mathbb{P}^{e'}(\mathbf{X} \mid Y)$

Can be learned from observed data

Confounding shift

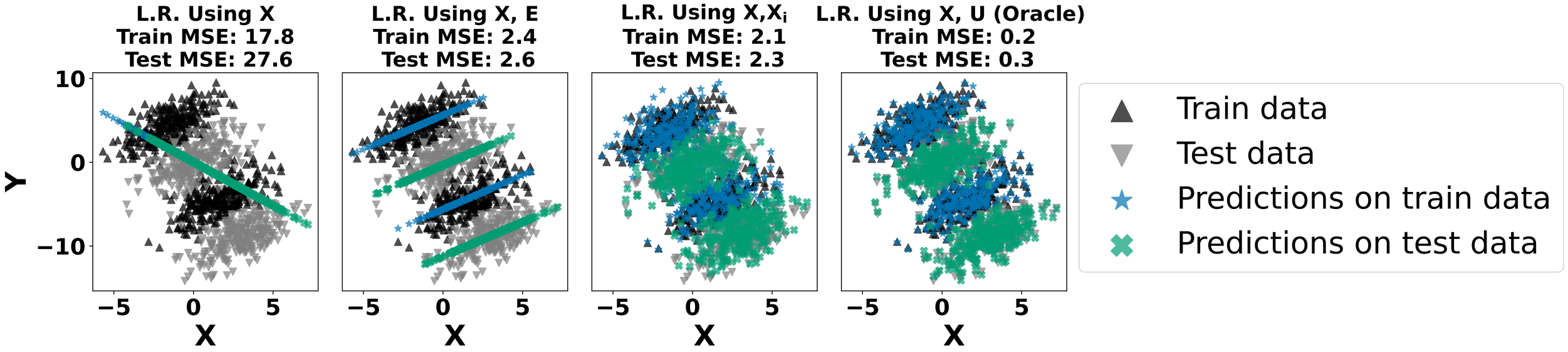


$$\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$$
$$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$$
Invariance: $\mathbb{P}^e(\mathbf{X}, Y \mid U) = \mathbb{P}^{e'}(\mathbf{X}, Y \mid U)$

Can't be learned from observed data

Motivation and Open Questions

- Predictors based only on <u>observed</u> covariates do not generalize



- Why do methods like MLP/Boosting generalize better than IRM-like methods?
- Why do **non-causal covariates** improve generalization performance?

Research Outcomes

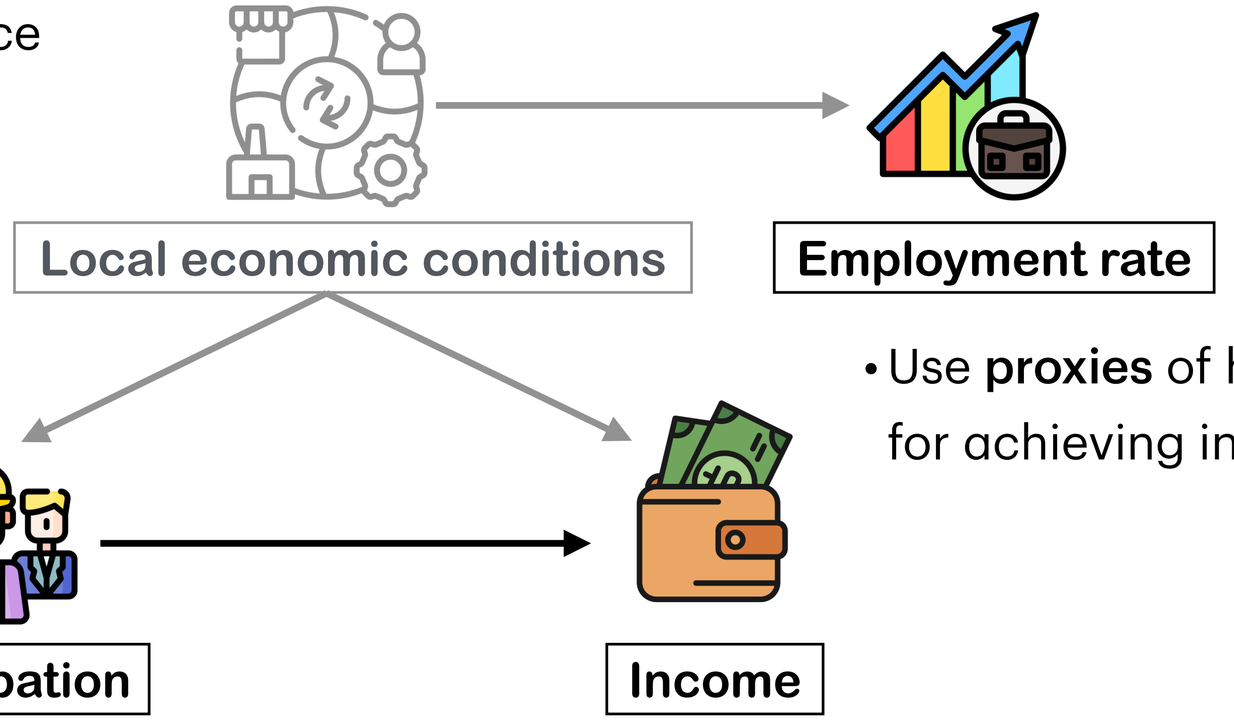

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- **Confounder-specific predictors** perform better than invariance learning methods[1]
- **Boosting** learns confounder-specific predictors[2]



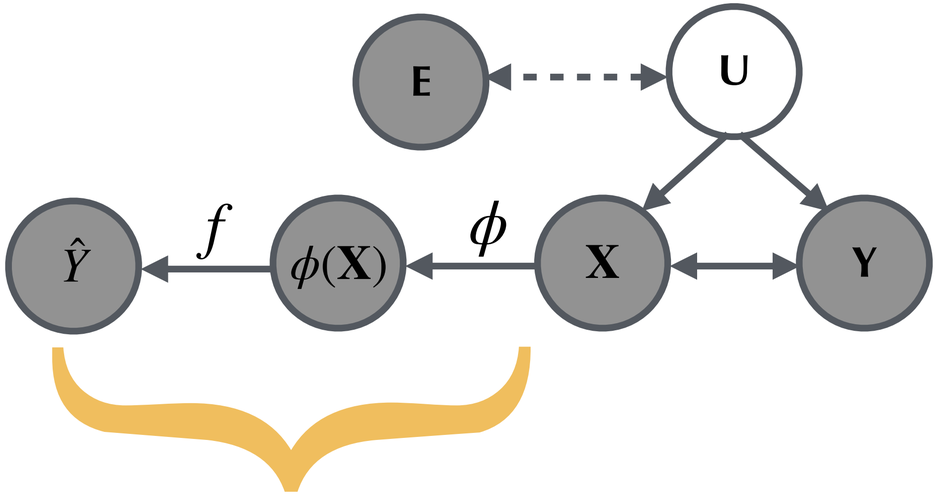
- Use **proxies** of hidden confounders for achieving invariance[1]

[1] Gowtham, Celia, Rebekka, Krikamol. **When Shift Happens — Confounding is to Blame**. To appear in ICLR 2026.

[2] Gowtham, Rajeev, Celia, Krikamol, Rebekka. **Boosting for Predictive Sufficiency**. To appear in ICLR 2026.

Causal Model

$\hat{Y} = (f \circ \phi)(\mathbf{X})$



Data Generating Process

- $\mathbf{X}$ is the set of covariates
- Y is the outcome
- U is a continuous (hidden) confounder
- E is a discrete environment variable
- E encodes a shift in $\mathbb{P}(U)$ across envs.

Computation Process

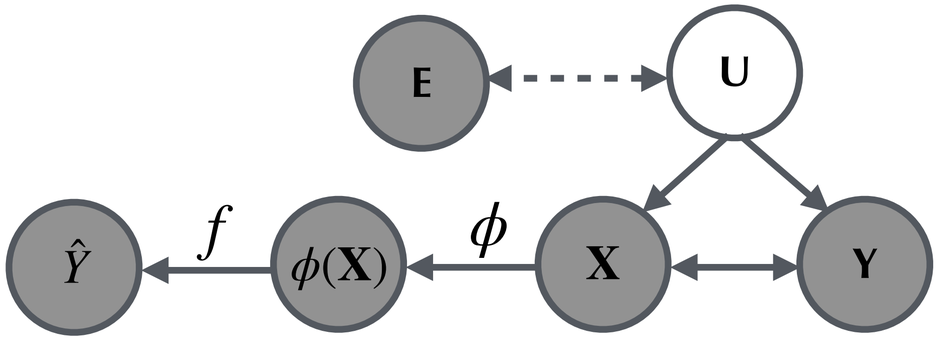
- $\phi(\mathbf{X})$ is a learned representation
- $\hat{Y}$ is the predicted outcome

Goal

Achieve high predictive information $I(Y; \hat{Y})$

Information-Theoretic Measures

$\hat{Y} = (f \circ \phi)(\mathbf{X})$



- Feature shift — $I(\phi(\mathbf{X}); E)$
- Label shift — $I(Y; E)$
- Concept shift — $I(Y; E \mid \phi(\mathbf{X}))$
- Variation — $I(\phi(\mathbf{X}); E \mid Y)$
- Conditional Informativeness — $I(\phi(\mathbf{X}); Y \mid E)$
- Residual — $I(\phi(\mathbf{X}); Y \mid \hat{Y})$

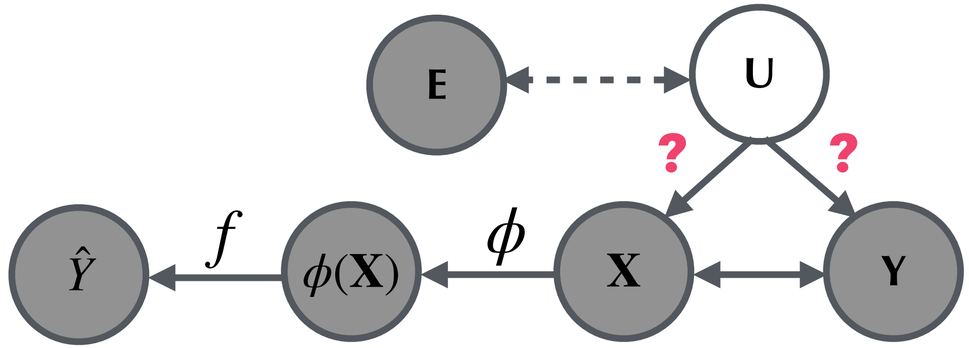
Earlier methods focus more on orange terms but not on blue terms
More on this later..

Decomposition of Predictive Information $I(Y; \hat{Y})$

Theorem-1

If we don't know (?) how U influences $\mathbf{X}$, Y and if $\mathbf{X} \leftrightarrow Y$, predictive information can be decomposed as follows:

$$\hat{Y} = (f \circ \phi)(\mathbf{X})$$



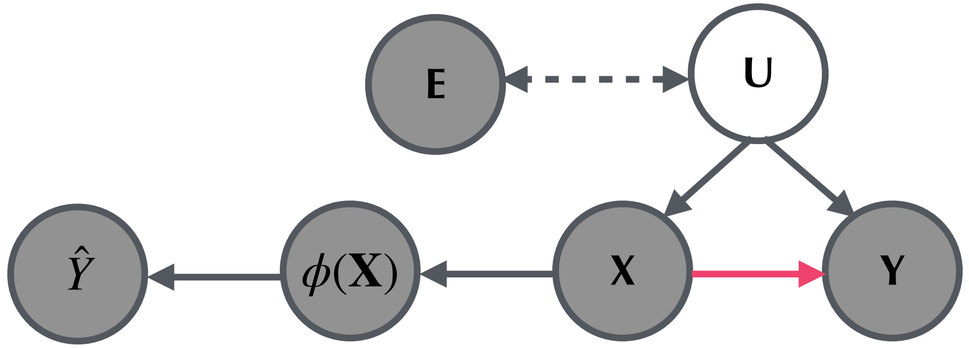
$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y \mid E)}^{\text{Cond. Inform.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E \mid Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E \mid \phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y \mid \hat{Y})}^{\text{Residual}}$$

Role of Different Terms in OOD Generalisation

$$I(Y;\hat{Y}) = \overbrace{I(\phi(\mathbf{X});Y\,|\,E)}^{\textbf{Cond. Inform.}} - \frac{\overbrace{I(\phi(\mathbf{X});E\,|\,Y)}^{\textbf{Variation}}}{2} + \frac{\overbrace{I(Y;E)}^{\textbf{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X});E)}^{\textbf{Feature shift}}}{2} - \frac{\overbrace{I(Y;E\,|\,\phi(\mathbf{X}))}^{\textbf{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X});Y\,|\,\hat{Y})}^{\textbf{Residual}}$$

- Maximizing **conditional informativeness** is good; potentially bad for fair predictions
- Minimizing **conditional informativeness** implies information loss
- Minimizing **residual** ensures all the information in $\phi(\mathbf{X})$ is used by the classifier f
- Fair predictions can be obtained by minimizing **feature shift** at the cost of predictive information
- Minimizing **variation** implies that, given Y, $\phi(\mathbf{X})$ should not change across environments
- **Concept shift** should be minimized, i.e., given $\phi(\mathbf{X})$, Y should not change across environments
- **Label shift** is a constant w.r.t. learning; there are methods that are robust to label shift

Special Case #1



U influences both $\mathbf{X}$, Y and $\mathbf{X} \rightarrow Y$

$$\phi(\mathbf{X}) \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$$

$$\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftrightarrow E$$

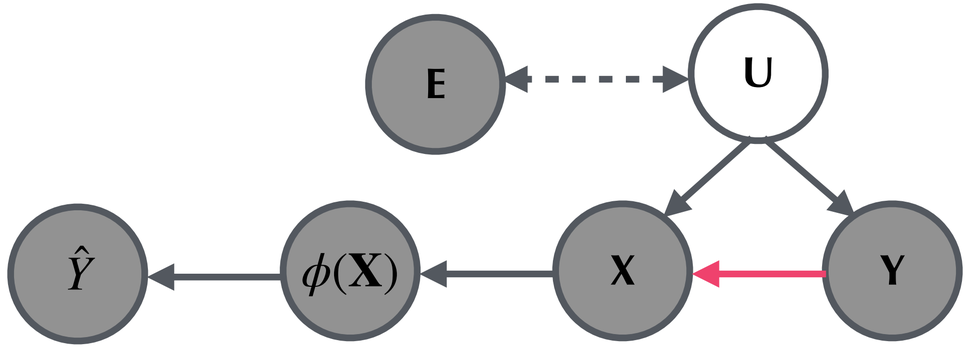
$$Y \leftarrow U \leftrightarrow E$$

$$Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$$

$$\overbrace{I(\phi(\mathbf{X}); E \mid Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}$$

$$\overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E \mid \phi(\mathbf{X}))}^{\text{Concept shift}}$$

Special Case #2



U influences both $\mathbf{X}$, Y and $Y \rightarrow \mathbf{X}$

$$\phi(\mathbf{X}) \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$$

$$\phi(\mathbf{X}) \leftarrow \mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$$

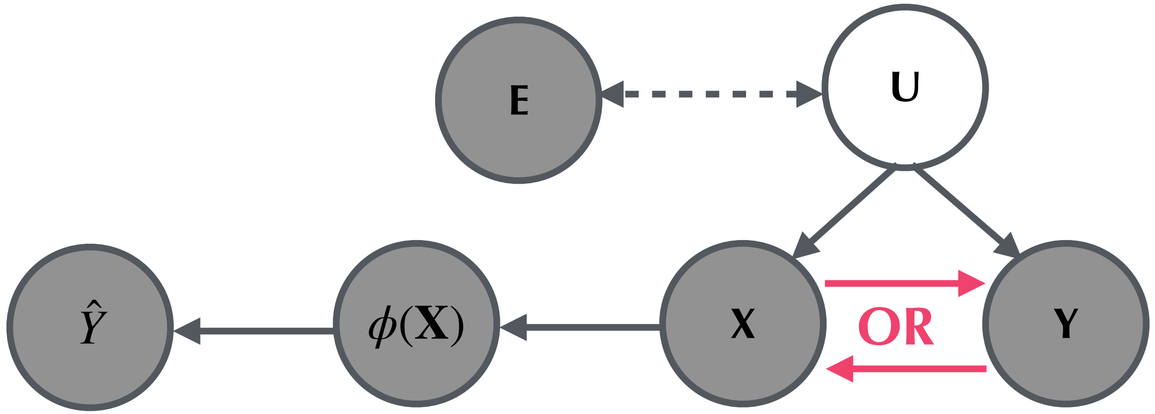
$$Y \leftarrow U \leftrightarrow E$$

$$Y \rightarrow \mathbf{X} \leftarrow U \leftrightarrow E$$

$$\overbrace{I(\phi(\mathbf{X}); E \mid Y)}^{\text{Variation}} \leq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}$$

$$\overbrace{I(Y; E)}^{\text{Label shift}} \leq \overbrace{I(Y; E \mid \phi(\mathbf{X}))}^{\text{Concept shift}}$$

Predictive Information Under Hidden Confounding Shift



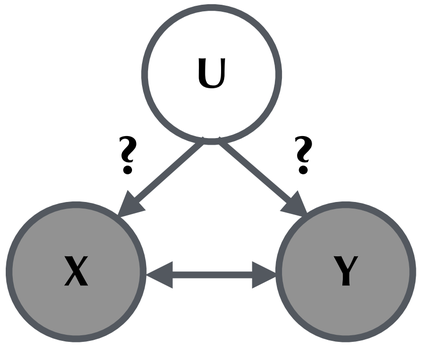
Theorem 2

If U influences both $\mathbf{X}, Y$ and if either (i) $\mathbf{X} \to Y$ or (ii) $Y \to \mathbf{X}$

$$I(Y;\hat{Y}) = \overbrace{I(\phi(\mathbf{X});Y\,|\,E)}^{\text{Cond. Inform.}} - \overbrace{I(\phi(\mathbf{X});Y\,|\,\hat{Y})}^{\text{Residual}}$$

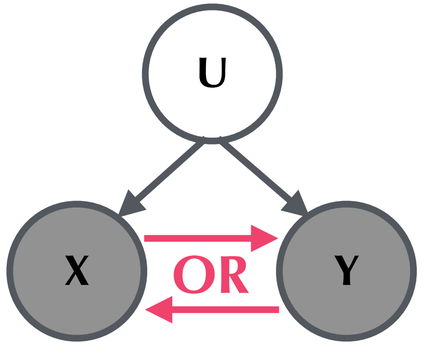
Summary of Theorem 1 & 2

If we don't know how U influences $\mathbf{X}$, Y and $\mathbf{X} \leftrightarrow Y$



$$I(Y;\hat{Y}) = \overbrace{I(\phi(\mathbf{X});Y\,|\,E)}^{\textbf{Cond. Inform.}} - \frac{\overbrace{I(\phi(\mathbf{X});E\,|\,Y)}^{\textbf{Variation}}}{2} + \frac{\overbrace{I(Y;E)}^{\textbf{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X});E)}^{\textbf{Feature shift}}}{2} - \frac{\overbrace{I(Y;E\,|\,\phi(\mathbf{X}))}^{\textbf{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X});Y\,|\,\hat{Y})}^{\textbf{Residual}}$$

If U influences both $\mathbf{X}$, Y and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$

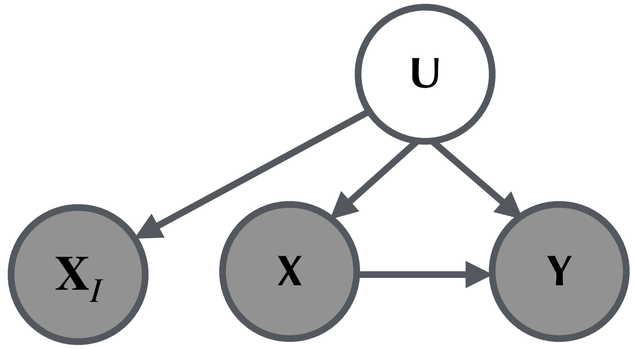


Most of real-world tabular data

$$I(Y;\hat{Y}) = \overbrace{I(\phi(\mathbf{X});Y\,|\,E)}^{\textbf{Cond. Inform.}} - \overbrace{I(\phi(\mathbf{X});Y\,|\,\hat{Y})}^{\textbf{Residual}}$$

Informative Covariates

Definition: $\mathbf{X}_I$ that are neither causal nor anti-causal are informative to Y if $\mathbf{X}_I \not\perp\!\!\!\perp Y \mid \mathbf{X}$



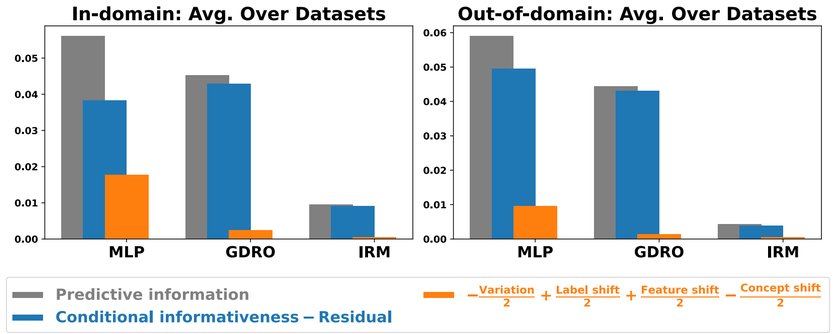
Proposition: Adding informative covariates to existing covariates

- $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y \mid E) > I(\phi_1(\mathbf{X}); Y \mid E)$ Conditional informativeness increases
- $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$ Feature shift increases
- $I(Y; E \mid \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E \mid \phi_1(\mathbf{X}))$ **Concept shift** decreases
- $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y)$ Variation increases

Experiments: Hidden Confounding Shift

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E\|Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E\|\mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004

Hidden confounding shift is common (as measured by a shift in observed variables)

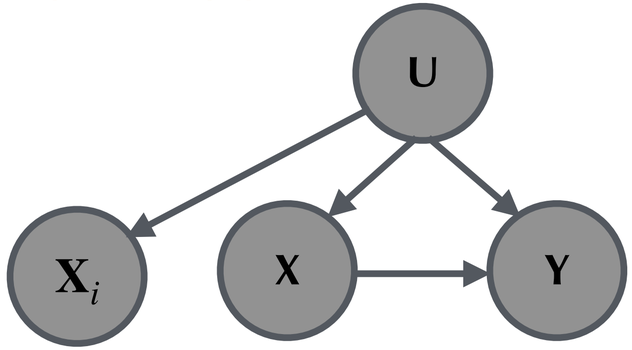


Method	ID Test Acc.	OOD Test
MLP	83.67	80.89
GDRO	82.48	79.64
IRM	67.65	65.48

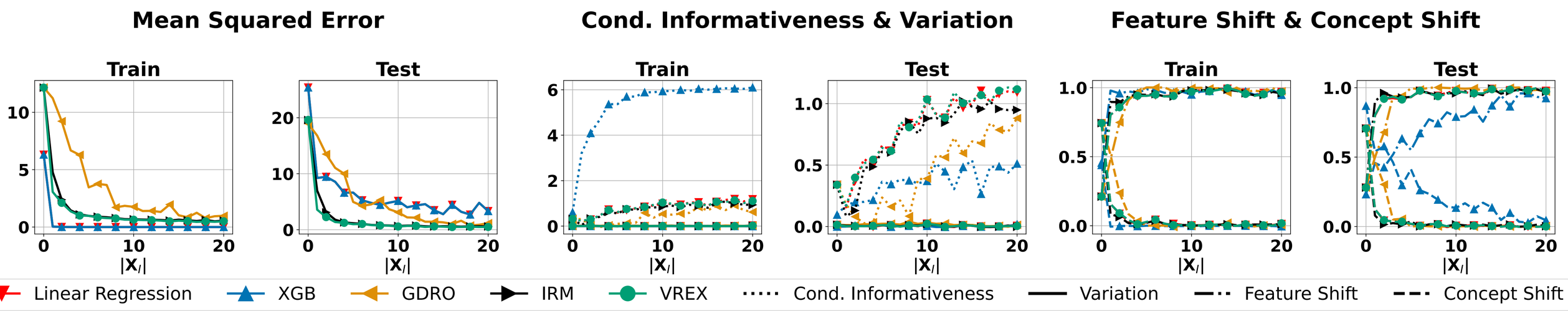
- Predictive information always correlates with accuracy
- Conditional informativeness - residual frequently correlated with accuracy
- Middle terms frequently go to zero

Synthetic Experiment: Informative Covariates

Causal (C) ⊆ Arguably Causal (AC) ⊆ All (A)

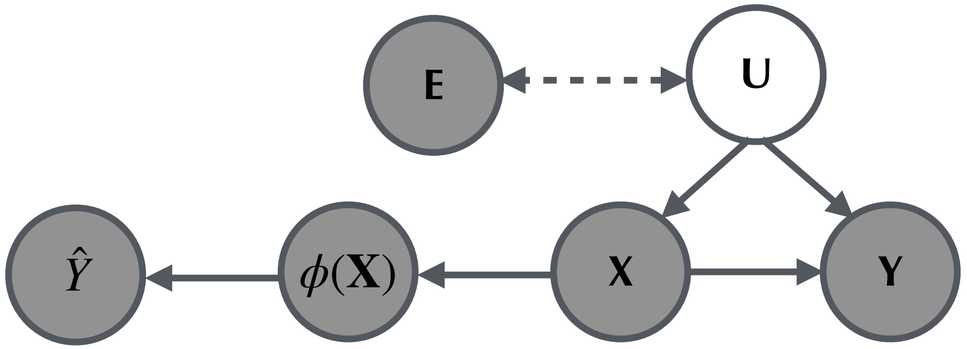


$$i \in \{1,\ldots,20\}$$



When shift happens, who is to trust?

Predictive Sufficiency



$$I(Y;\hat{Y}) = \overbrace{I(\phi(\mathbf{X});Y\,|\,E)}^{\textbf{Cond. Inform.}} - \overbrace{I(\phi(\mathbf{X});Y\,|\,\hat{Y})}^{\textbf{Residual}}$$

α - predictive sufficiency $\implies$ $I(Y-\hat{Y};E\mid\hat{Y}) \le \alpha$

Predictive sufficiency and predictive information are related as:

$$I(Y-\hat{Y};E\mid\hat{Y}) = -I(Y;\phi(\mathbf{X})\mid E,\hat{Y}) + I(Y;\phi(\mathbf{X})\mid\hat{Y}) + I(Y;E\mid\phi(\mathbf{X}))$$

Research question: can Boosting achieve α-predictive sufficiency?

Boosting for Predictive Sufficiency

Theorem

$\exists \;\; T < \infty$ such that $\hat{Y}_t$ learned by a boosting algorithm after $t \geq T$ rounds is α-predictive sufficient.

$$T = \frac{H(Y) - H(Y \mid \mathbf{X}, E) - \alpha - I(Y; \hat{Y}_0)}{p \cdot \gamma}$$

If E is unknown:

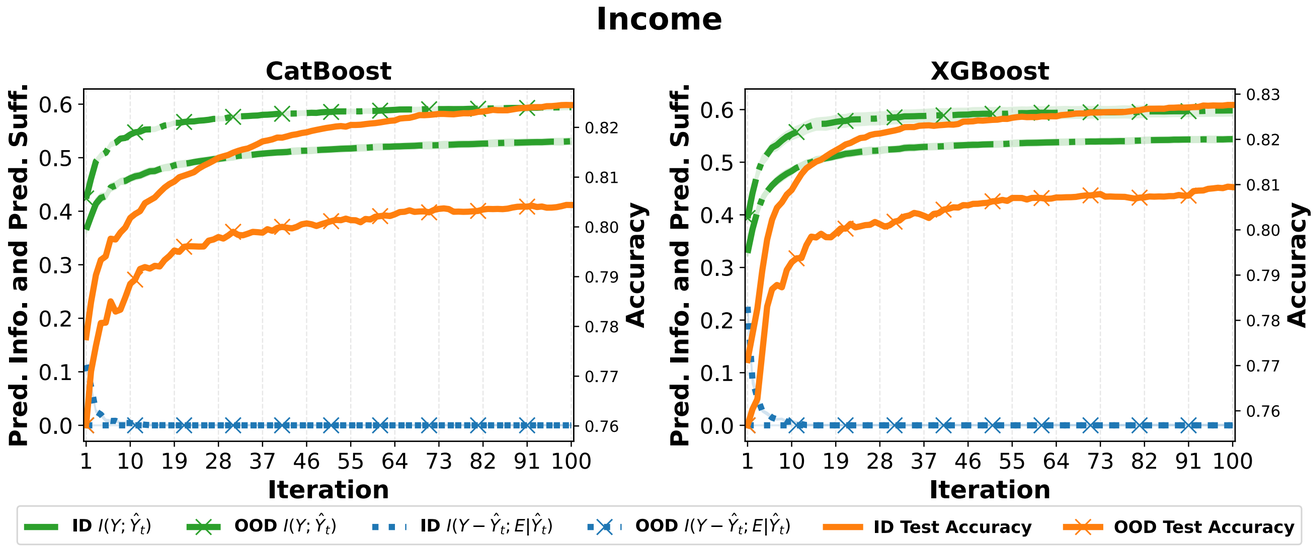
$$T = \frac{H(Y) - \alpha - I(Y; \hat{Y}_0)}{p \cdot \gamma}$$

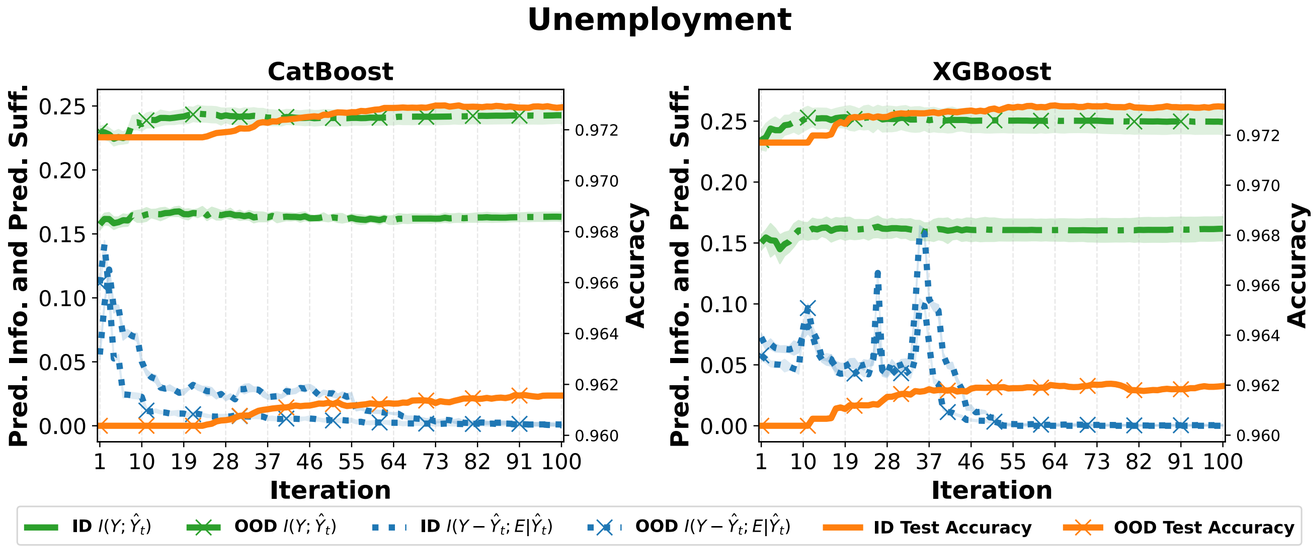
Corollary

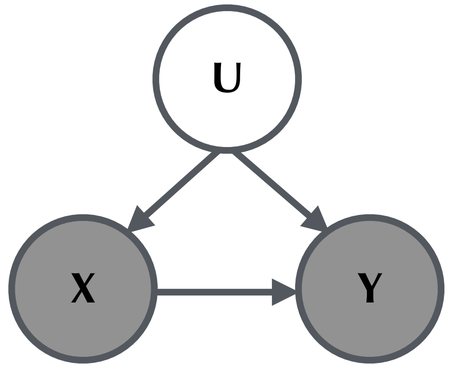
$\exists \;\; T < \infty$ such that $\hat{Y}_t$ learned by a boosting algorithm after $t \geq T$ rounds satisfies $H(U \mid \hat{Y}_t) \leq \delta$

$$T = \frac{H(U) - \delta - c \cdot I(Y; \hat{Y}_0)}{c \cdot p \cdot \gamma}$$

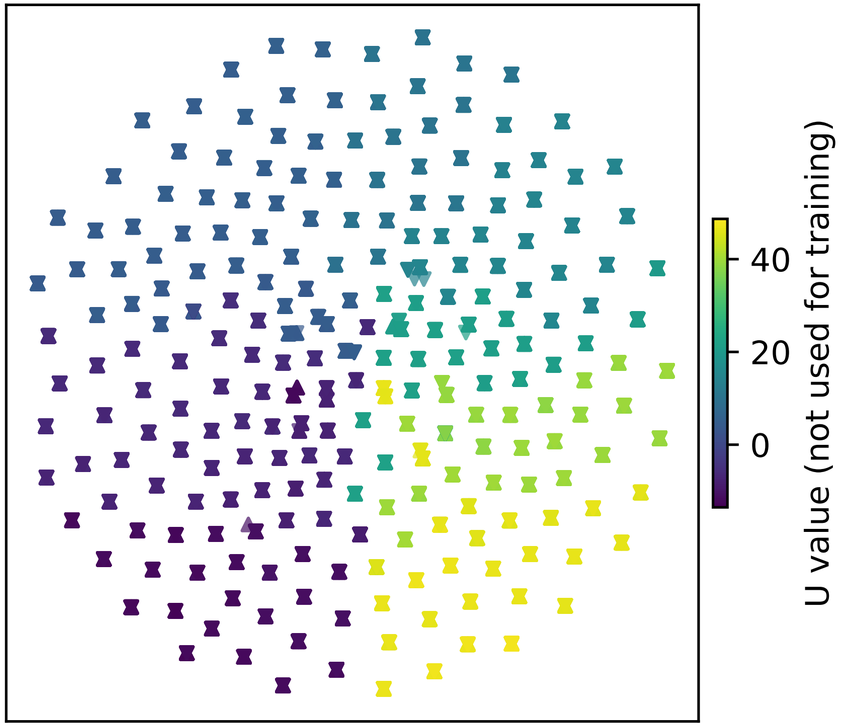
Synthetic Experiments







CatBoost (#estimators=100, max depth=10)
ID MSE=2.495, OOD MSE=2.702
ARI(U,$\phi(X)$)=0.730, NMI(U,$\phi(X)$)=0.842



Conclusion

- When shift happens, **confounding** is to blame.
- Adding **unjustified assumptions** can comes at the cost of generalisability.
- Addressing confounding shifts requires predictors with **epistemic humility,** enabling robustness to unknown unknowns.

References



ICLR
International Conference On Learning Representations

WHEN SHIFT HAPPENS - CONFOUNDING IS TO BLAME

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ABSTRACT

Distribution shifts introduce uncertainty that undermines the robustness and generalization capabilities of machine learning models. While conventional wisdom suggests that learning causal-invariant representations enhances robustness to such shifts, recent empirical studies present a counterintuitive finding: (i) empirical risk minimization (ERM) can rival or even outperform state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) OOD generalization performance improves when all available covariates, including non-causal ones, are utilized. We present theoretical and empirical explanations that attribute this phenomenon to hidden confounding. Shifts in hidden confounding induce changes in data distributions that violate assumptions commonly made by existing approaches. Under such conditions, we prove that generalization requires learning environment-specific relationships, rather than relying solely on invariant ones. Furthermore, we explain why models augmented with non-causal but informative covariates can mitigate the challenges posed by hidden confounding shifts. These findings offer new theoretical insights and practical guidance, serving as a roadmap for future research on OOD generalization and principled covariate-selection strategies.

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BOOSTING FOR PREDICTIVE SUFFICIENCY

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[†]CISPA Helmholtz Center for Information Security [‡]University of Amsterdam

ABSTRACT

Out-of-distribution (OOD) generalization is a defining hallmark of truly robust and reliable machine learning systems. Recently, it has been empirically observed that existing OOD generalization methods often underperform on real-world tabular data, where hidden confounding shifts drive distribution shifts that boosting models handle more effectively. Earlier work attributes a part of boosting's success to variance reduction, handling missing covariates, feature selection, and connections to multicalibration. Complementary to these explanations, we uncover a crucial reason behind boosting's success in OOD generalization: its ability to identify environments created by hidden confounding shifts and maximize predictive performance within those environments. To this end, this paper introduces an information-theoretic notion called α-predictive sufficiency and formalizes its connection to OOD generalization under hidden confounding shift. We show that boosting implicitly identifies suitable environments and produces an α-predictive sufficient predictor. We validate our theoretical results through synthetic and real-world experiments and show that boosting achieves robust performance by identifying these environments and maximizing the mutual information between predictions and true outcomes.

